

CPE 381: Fundamentals of Signals and Systems for Computer Engineers

03 Continuous-Time Signals and Systems

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Outline

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- 2.2 Basic Signals as Building Blocks
- 2.3 Modulation and Windowing
- 2.4 Linearity of Signals

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- 3.2 Convolutional Integral
- 3.3 Causality and Stability



Motivation

Signals and Systems is 'Grandfather' of Data Science for Electrical and Computer Engineers

Continuous-Time Signals



0	0	0	0	1	0	1	0	0	0	0	1	0	1	0	1	0	0	0
1	1	0	1	1	0	1	0	1	1	1	0	0	0	1	0	0	0	0
0	0	0	0	1	1	1	0	0	0	1	1	0	0	0	0	1	1	1

Classification of Signals

We care about the following properties when dealing with signals:

- ⚡ Predictability: Random or Deterministic
- ⚡ Variations of time and amplitude: continuous, discrete (time or x-axis) / quantized (amplitude or y-axis)
- ⚡ Periodic/Aperiodic
- ⚡ Finite energy/finite power; Infinite energy/Infinite power

Continuous-Time Signals

2.1

Operation on Signals



0	1	0	1	0	1	1	0	1	1	0	1	0	0	0
1	1	1	1	1	0	1	1	1	0	0	1	1	1	1

Basic Mathematical Operations

- ⚡ Addition: $x(t) + y(t)$
- ⚡ Subtraction: $x(t) - y(t)$
- ⚡ Constant multiplication: $kx(t)$ where k is a constant

Time-shift

⚡ $x(t - \tau) \rightarrow$ Signal is delayed

⚡ $x(t + \tau) \rightarrow$ Signal is advanced

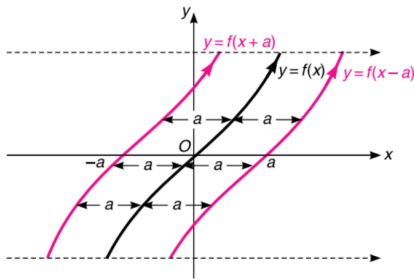
$f(x)$ transforms to $f(x - a)$

i.e., $f(x) \longrightarrow f(x - a)$; a is positive. Shift the graph of $f(x)$ through ' a ' unit towards right

$f(x)$ transforms to $f(x + a)$.

i.e., $f(x) \longrightarrow f(x + a)$; a is positive. Shift the graph of $f(x)$ through ' a ' units towards left.

Graphically it could be stated as



Time Reflection

Note: The book doesn't specify whether to take the mirror image along the y-axis or not and it is confusing because the signal used in example 1.3.1 is symmetric with respect to both the x and y axes.

$f(x)$ transforms to $f(-x)$

i.e.,

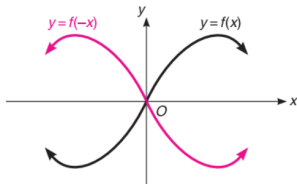
$$f(x) \longrightarrow f(-x)$$

To draw $y = f(-x)$, take the image of the curve $y = f(x)$ in y-axis as plane mirror.

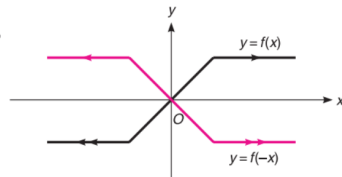
OR

“Turn the graph of $f(x)$ by 180° about y-axis.”

Graphically it is stated as;



OR



⚡ $x(t) \rightarrow x(-t)$: take mirror image **along the y-axis**

Signal Stretching along y-axis

⚡ $f(x) \rightarrow af(x)$; $a > 1$: Stretch the graph of $f(x)$ 'a' times along y-axis.

⚡ $f(x) \rightarrow \frac{1}{a}f(x)$; $a > 1$: Shrink the graph of $f(x)$ 'a' times along y-axis.

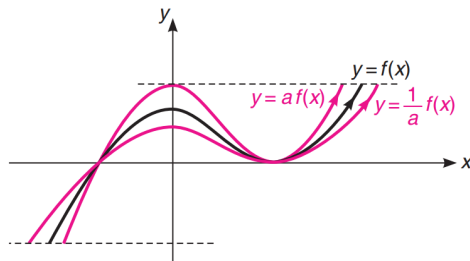
$f(x)$ transforms to $a f(x)$

i.e., $f(x) \longrightarrow af(x)$; $a > 1$

Stretch the graph of $f(x)$ 'a' times along y-axis.

$$f(x) \longrightarrow \frac{1}{a}f(x); a > 1.$$

Shrink the graph of $f(x)$ 'a' times along y-axis.



Signal Stretching along x-axis

⚡ $f(x) \rightarrow f(ax); \quad a > 1$: Stretch the graph of $f(x)$ ' a ' times along x-axis.

⚡ $f(x) \rightarrow f\left(\frac{1}{a}x\right); \quad a > 1$: Shrink the graph of $f(x)$ ' a ' times along x-axis.

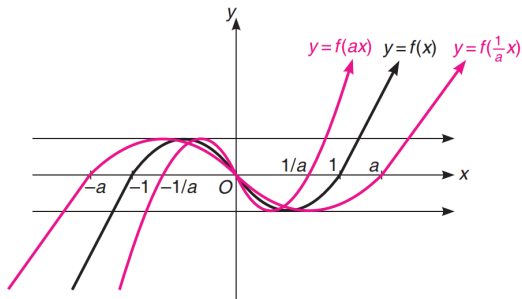
$f(x)$ transforms to $f(ax)$

i.e., $f(x) \longrightarrow f(ax); \quad a > 1$

Shrink (or contract) the graph of $f(x)$ ' a ' times along x-axis.

again $f(x) \longrightarrow f\left(\frac{1}{a}x\right); \quad a > 1$

Stretch (or expand) the graph of $f(x)$ ' a ' times along x-axis.

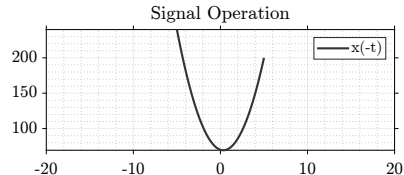
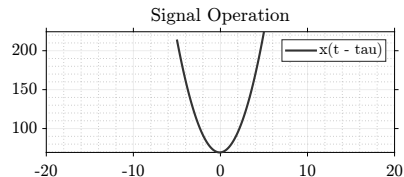
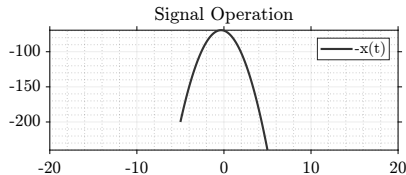
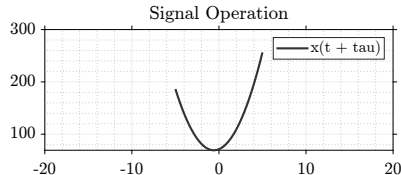
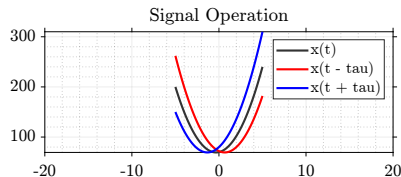


Example and MATLAB Code

$$x(t) = 6t^2 + 4t + 70$$

Code:

https://github.com/rahulbhadani/CPE381_FA26/blob/main/Code/Ch03_signal_operation.m



Even and Odd Signals

⚡ Even Signal: $x(t) = x(-t)$

⚡ Odd Signal: $x(t) = -x(-t)$

⚡ Any signal can be represented by the sum of even and odd signals

$$y(t) = y_e(t) + y_o(t)$$

$$y_e(t) = 0.5[y(t) + y(-t)]$$

$$y_o(t) = 0.5[y(t) - y(-t)]$$

Periodic Signals

- ⚡ Defined for all possible values of t , $-\infty < t < \infty$.
- ⚡ There is the real value $T_0 \in \mathbb{R}^+$, called the fundamental frequency such that $x(t + kT_0) = x(t)$, $k \in \mathbb{I}$.
- ⚡ A constant signal is periodic of a non-definable fundamental period.
- ⚡ $A \cos(\omega t + \theta)$, $\omega = 2\pi/T_0$, $\omega = 2$, $\theta = -\pi/2$, $A = 2$.

What's the fundamental frequency, $1/T_0$?

Energy and Power of Signals

⚡ Energy:

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

⚡ Power:

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$$

A signal is called finite power if the signal power is finite.

Basic Signals as Building Blocks



1	0	0	0	0	0	1	0	0	0	0	0	0	1	1
0	1	1	0	1	0	1	1	0	1	0	1	1	0	0

Complex Exponentials

Consider $A = |A|e^{j\theta}$, $a = r + k\Omega_0$

⚡ $x(t) = Ae^{at} = \dots$

⚡ Real part $f(t) = \operatorname{Re}\{x(t)\}, = \dots$

$-|A|e^{rt} \leq f(t) \leq |A|e^{rt}$. $r < 0$, $f(t)$ is damped, $r > 0$, $f(t)$ grows.

⚡ Imaginary part $g(t) = \operatorname{Im}\{x(t)\}, = \dots$

Sinusoids

A sinusoid of the general form:

$$A \cos(\Omega_0 t + \theta) = A \sin(\Omega_0 t + \theta + \pi/2), \quad -\infty < t < \infty$$

- ⚡ A is the amplitude
- ⚡ $\Omega_0 = 2\pi f_0$ is angular frequency in rad/s.
- ⚡ θ is phase shift
- ⚡ Fundamental period T_0 is

$$T_0 = \frac{2\pi}{\Omega_0} = \frac{1}{f_0}$$

Rectangular pulse and Unit impulse

⚡ A rectangular pulse of duration Δ and unit area:

$$p_{\Delta}(t) = \begin{cases} \frac{1}{\Delta}, & -\Delta/2 \leq t \leq \Delta/2 \\ 0, & \text{otherwise} \end{cases}$$

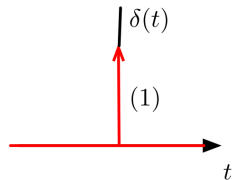
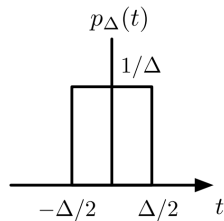
⚡ Unit Impulse:

$$\delta(t) = \lim_{\Delta \rightarrow 0} p_{\Delta}(t)$$

⚡ Calculate

$$\int_{-\infty}^t p_{\Delta}(t) dt$$

Hint: there are three possible cases.



Unit Step

⚡ Integration of rectangular pulse:

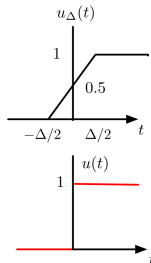
$$u_{\Delta}(t) = \int_{-\infty}^t p_{\Delta}(t) = \begin{cases} 1, & t \geq \frac{\Delta}{2} \\ \frac{1}{\Delta}(t + \frac{\Delta}{2}), & -\frac{\Delta}{2} \leq t \leq \frac{\Delta}{2} \\ 0, & t < -\frac{\Delta}{2} \end{cases}$$

⚡ Limit case:

$$\lim_{\Delta \rightarrow 0} u_{\Delta}(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

⚡ A common case is to ignore $t = 0$ case, which gives us unit step function as

$$u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$



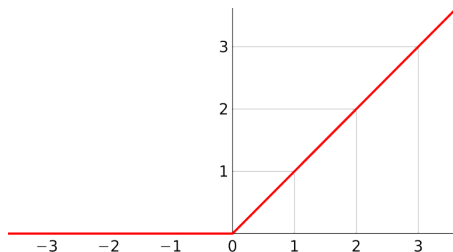
Ramp Signal

The ramp signal is $r(t) = tu(t)$

- ⚡ The relation between the ramp, the unit step, and the unit impulse:

$$\frac{dr(t)}{dt} = u(t)$$

$$\frac{d^2r(t)}{dt^2} = \delta(t)$$



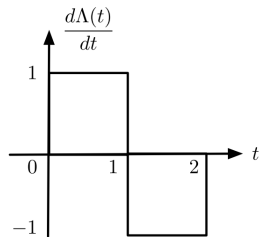
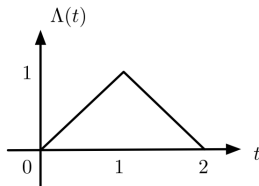
Triangular Pulse

The triangular pulse is

$$\Lambda(t) = \begin{cases} t, & 0 \leq t \leq 1 \\ -t + 2, & 1 < t \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

⚡ $\Lambda(t)$ can also be written as:

$$\Lambda(t) = r(t) - 2r(t - 1) + r(t - 2)$$



Triangular Pulse as Ramp Functions I

Let's verify this by evaluating $\Lambda(t)$ at different intervals:

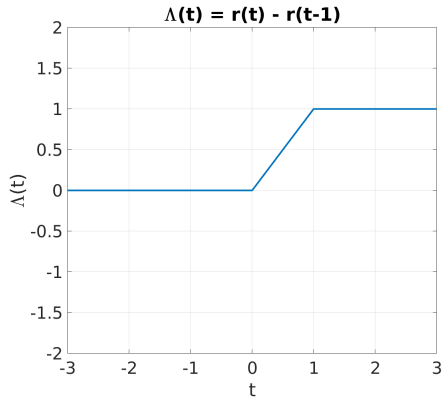
First, find out the first part:

$$\Lambda_1(t) = \begin{cases} t, & 0 \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

which can be written as:

can be written using the ramp function $r(t)$ as:

$$\Lambda_1(t) = r(t) - r(t - 1)$$



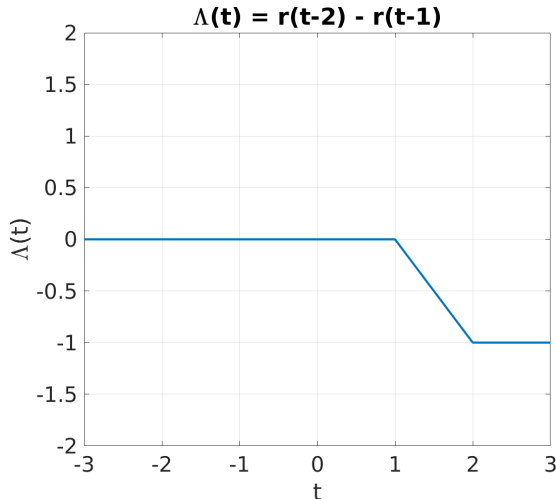
Triangular Pulse as Ramp Functions II

Second part:

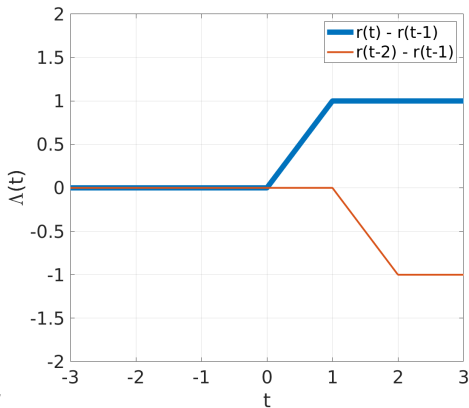
$$\Lambda_1(t) = \begin{cases} -t + 2, & 1 < t \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

can be written using the ramp function $r(t)$ as:

$$\Lambda_1(t) = r(t - 2) - r(t - 1)$$



Triangular Pulse as Ramp Functions III



Putting together

Code for the graph:

https://github.com/rahulbhadani/CPE381_FA26/blob/main/Code/Ch03_ramp.m

Sifting Property

The product of $f(t)$ and $\delta(t)$ gives zero everywhere except at the origin where we get an impulse of area $f(0)$, that is, $f(t)\delta(t) = f(0)\delta(t)$.

Hence,

$$\int_{-\infty}^{\infty} f(t)\delta(t)dt = \int_{-\infty}^{\infty} f(0)\delta(t)dt = f(0) \int_{-\infty}^{\infty} \delta(t)dt = f(0)$$

This is called **Sifting Property**.

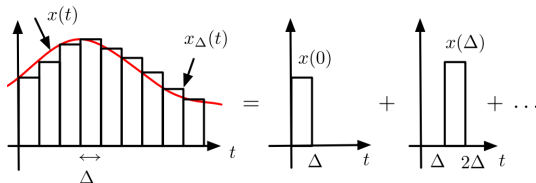
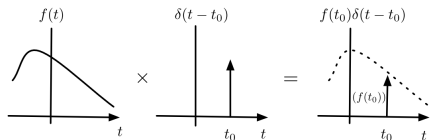
If we delay or advance the $\delta(t)$ function in the integrated, the result is that all values of $f(t)$ are sifted out except for the value corresponding to the location of the delta function, that is,

$$\int_{-\infty}^{\infty} f(t)\delta(t - \tau)dt = \int_{-\infty}^{\infty} f(\tau)\delta(t - \tau)dt = f(\tau) \int_{-\infty}^{\infty} \delta(t - \tau)dt = f(\tau) \quad \text{for any } \tau$$

Generic Representation of Signals

Hence, if we do integration in terms of variable τ , we get a generic representation of signals in terms of impulse and shifted impulse.

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$$



⚡ Approximation of $x(t)$:

$$x_{\Delta}(t) = \sum_{-\infty}^{\infty} x_{\Delta}(t - k\Delta) = \sum_{-\infty}^{\infty} x(k\Delta) p_{\Delta}(t - k\Delta) \Delta$$

In the limit as $\Delta \rightarrow 0$ these pulses become impulses, separated by an infinitesimal value:

$$\lim_{\Delta \rightarrow 0} x_{\Delta}(t) \rightarrow x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$$

Modulation and Windowing



1	1	0	0	0	1	1	1	0	0	1	0	0	1	0
0	1	0	1	1	0	1	1	0	1	0	0	0	0	0

Modulation

Multiplication by a complex exponential shifts the frequency of the original signal.

Definition

Superimposing a low-frequency signal on a high-frequency carrier signal is called **Modulation**.

Example:

Consider an exponential signal $x(t) = e^{j\Omega_0 t}$ of frequency Ω_0 . If we multiply an exponential $e^{j\phi t}$ with $x(t)$, then:

$$x(t)e^{j\phi t} = e^{j(\Omega_0 + \phi)t} = \cos((\Omega_0 + \phi)t) + j \sin((\Omega_0 + \phi)t)$$

$\phi > 0$: the frequency of new exponential is greater than Ω_0 , otherwise lower.

Various Types of Modulation

$$A(t)\cos(\Omega(t)t + \theta(t))$$

- ⚡ $A(t)$ changes: Amplitude Modulation
- ⚡ $\Omega(t)$ changes: Frequency Modulation
- ⚡ $\theta(t)$ changes: Phase Modulation

Windowing

For a window signal $w(t)$, the time-windowed signal

$x(t)w(t)$ displays $x(t)$ within the support of $w(t)$.

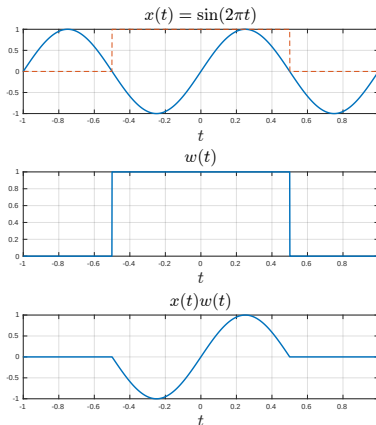
Example:

$$x(t) = \sin(2\pi t)$$

$$w(t) = \begin{cases} 1 & \text{if } -0.5 \leq t \leq 0.5 \\ 0 & \text{otherwise} \end{cases}$$

Code for the graph:

https://github.com/rahulbhadani/CPE381_FA26/blob/main/Code/Ch03_windowing.m



Classwork

Classwork

Classwork

Classwork

Classwork

Classwork

Classwork

Classwork

Classwork

Continuous-Time Signals

2.4

Linearity of Signals



0	0	1	1	0	1	1	0	1	1	0	1	0	1	1
0	1	0	1	0	0	0	0	1	0	0	0	1	1	1

Properties of Linearity

A signal f is linear if it has these properties,

- 1 Homogeneity (scaling): $f(at) = af(t)$ where a is a constant scalar.
- 2 Additivity: $f(t_1 + t_2) = f(t_1) + f(t_2)$.

The homogeneity and additivity together are called the **principle of superposition**

Example 1

Consider a signal $f(t) = 5t$. Does it satisfy homogeneity?

Solution:

- 1 Apply the function to a scaled input, i.e. $a \cdot t$:

$$f(a \cdot t) = 5(a \cdot t) = 5at$$

- 2 Now, scale the output $f(t)$ by the same factor a :

$$a \cdot f(t) = a \cdot (5t) = 5at$$

- 3 LHS = RHS. Hence it satisfies homogeneity.

Example 2

Consider a signal $f(t) = 5t$. Does it satisfy additivity?

Solution:

- 1 Apply the function to the sum of two inputs t_1 and t_2 :

$$f(t_1 + t_2) = 5(t_1 + t_2) = 5t_1 + 5t_2$$

- 2 Now, add the outputs of the function applied to each input individually:

$$f(t_1) + f(t_2) = (5t_1) + (5t_2) = 5t_1 + 5t_2$$

- 3 LHS = RHS. Hence it satisfies additivity.

Example 3

Consider a signal $f(t) = 5t + 3$. Does it satisfy homogeneity?

Example 4

Consider a signal $f(t) = t^2$. Does it satisfy additivity?

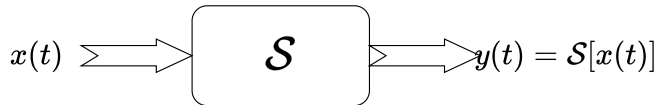
Continuous-Time Systems

1	1	1	0	0	1	0	1	1	1	1	0	1	1	0	0	1	0	0	0
0	0	0	0	0	1	1	0	0	0	1	0	0	0	1	1	1	0	0	1
0	1	1	0	1	1	0	0	1	1	1	0	1	0	1	0	1	0	0	0

System Thinking

System is an abstraction.

Think of a device or a process as a system – something that transforms one signal into another.



Continuous-time Systems

When inputs and outputs are continuous, the system is continuous.

Continuous-Time Systems

3.1

|

Linear Time-Invariant (LTI) Continuous-Time Systems



1	1	0	1	1	0	0	1	1	1	0	0	0	1	1
0	1	0	1	0	1	1	0	1	1	1	0	1	0	0

Linear Time Invariant (LTI) Continuous-Time Systems

For most analyses at system-level, we care about the following properties:

- ⚡ Linearity
- ⚡ Time-invariance
- ⚡ Causality
- ⚡ Stability

Linearity of a System

$$\mathcal{S}[a \cdot x(t) + b \cdot y(t)] = \mathcal{S}[a \cdot x(t)] + \mathcal{S}[b \cdot y(t)] = a \cdot \mathcal{S}[x(t)] + b \cdot \mathcal{S}[y(t)]$$

Time-invariance of a System

If

$$x(t) \Rightarrow y(t) = \mathcal{S}[x(t)],$$

then, the time-invariance says:

$$x(t \mp \tau) \Rightarrow y(t \mp \tau) = \mathcal{S}[x(t \mp \tau)].$$

Linear Time Invariant (LTI) Systems

A system that satisfies both linearity and time-invariance is an LTI system.

Class Problem 1

Consider a system $g(t) = \mathcal{S}[f(t)] = a[f(t)]^2 - b[f(t)]$

⚡ Is this system linear?

⚡ Is this system time-invariant?

Class Problem 2

Consider

$$g(\gamma) = \mathcal{S}[f(t)] = \int_{-\infty}^{\infty} f(t) \text{rect}(\gamma - t) dt$$

- ⚡ Is this system linear?
- ⚡ Is this system time-invariant?

Static Systems vs Dynamic Systems

In static systems, output depends on the current input but not on the historical input or output

In dynamic systems, output depends on current input, past inputs, and past outputs.

Differential equations are used to represent dynamic systems (literature also refers to them as dynamical systems).

Representation of Dynamic Systems

Dynamic systems are represented by ordinary differential equations (ODE) with constant coefficients:

$$a_n \frac{d^n \gamma(t)}{dt^n} + a_{n-1} \frac{d^{n-1} \gamma(t)}{dt^{n-1}} + \cdots + a_1 \frac{d\gamma(t)}{dt} + a_0 \gamma(t) = \\ b_m \frac{d^m x(t)}{dt^m} + b_{m-1} \frac{d^{m-1} x(t)}{dt^{m-1}} + \cdots + b_1 \frac{dx(t)}{dt} + b_0 x(t)$$

with n initial conditions, $x(t) = 0$ for $t < 0$.

This equation represents a linear ordinary differential equation with constant coefficients, where $\gamma(t)$ is the output, $x(t)$ is the input, and a_i and b_i are the constant coefficients.

Response of a Dynamic Systems

Assessing the output of the system:

- ⚡ Consider output due to input only, considering initial conditions to be zero. (**zero-state response**).
- ⚡ Consider output due to initial conditions only, considering inputs to be zero. (**zero-input response**).
- ⚡ Complete response is the sum of the above two.

Differential Equations for Ideal Systems

Electrical Inductance

$$v_{21} = L \frac{di}{dt}$$



Electrical Capacitance

$$i = C \frac{dv_{21}}{dt}$$



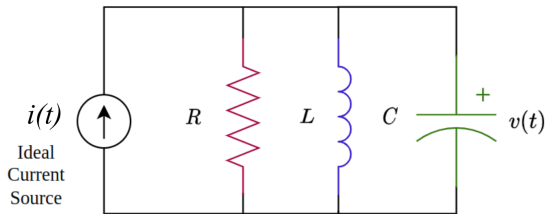
Electrical Resistance

$$i = \frac{v_{21}}{R}$$



Parallel RLC Circuit

$$\frac{v(t)}{R} + C \frac{dv(t)}{dt} + \frac{1}{L} \int_0^t v(t) dt = i(t)$$



Damped Mass-spring Oscillator

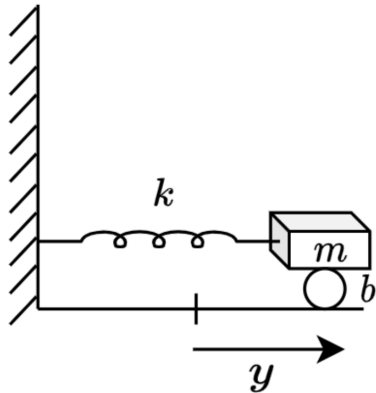
k = stiffness of the spring

b = damping coefficient

m = mass of the block

$y = y(t)$ time-dependent displacement from Equilibrium point.

$$m\ddot{y} = F - ky - b\dot{y}$$



Continuous-Time Systems

3.2

Convolutional Integral



○	○	1	○	○	1	○	1	1	○	1	1	1	○	1
○	○	○	○	1	○	1	○	1	1	1	1	1	1	○

Response to an Impulse Signal

Output when a system is presented with a unit-impulse signal is called **response to an impulse signal**, or in short **impulse response**.

Why is Impulse Response Useful?

It allows us to predict what the system's output will look like in the **time-domain**.

Do you remember generalized signal representation using impulse signals?

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$$

What would we get if we pass $x(t)$ through an LTI system $\mathcal{S}$?

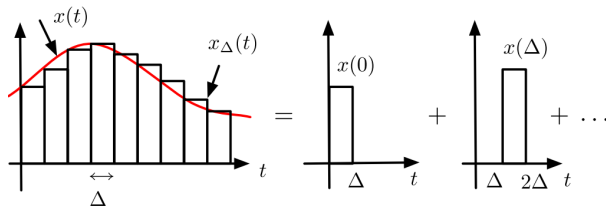
Impulse Response

What would we get if we pass $x(t)$ through an LTI system $\mathcal{S}$?

We get

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau$$

where $h(t)$ is the impulse response. Out is the superposition of the responses of each term in the figure.



Definition of Convolution Integral

The response of an LTI system $\mathcal{S}$, represented by its impulse response $h(t)$, to any signal $x(t)$ is the **convolution integral**.

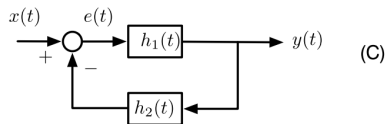
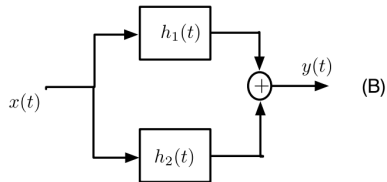
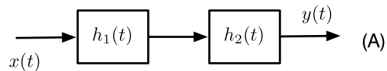
$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau = \int_{-\infty}^{\infty} x(t - \tau)h(\tau)d\tau = [x * h](t) = [h * x](t)$$

where $*$ stands for the convolution integral of the input signal $x(t)$ and the impulse response $h(t)$ of the system.

Some Points on Convolution Integral

- ⚡ Impulse response is fundamental to characterizing an LTI system.
- ⚡ A system represented by the convolution integral is LTI. It is a general representation of an LTI system.

Putting Multiple Systems Together



$$\text{A: } h(t) = [h_1 * h_2](t) = [h_2 * h_1](t)$$

$$\text{B: } h(t) = h_1(t) + h_2(t)$$

$$\text{C: } h(t) = [h_1 - h * h_1 * h_2](t)$$

Continuous-Time Systems

3.3

Causality and Stability



1	1	1	1	0	0	1	1	1	1	1	1	1	0	1
0	1	0	0	1	0	0	1	0	0	1	1	1	0	1

Causality

⚡ Cause-effect relationship,

⚡ A system is causal if

- whenever its input $x(t) = 0$, and there is no initial condition, $y(t) = 0$.
- the output $y(t)$ doesn't depend on the future inputs.

⚡ Causality is independent of linearity and time-invariance.

LTI system with impulse response $h(t)$ is causal if $h(t) = 0$ for $t < 0$, and

$$y(t) = \int_0^t x(\tau)h(t - \tau)d\tau$$

Bounded Input Bounded Output Stability

⚡ If input is bounded, i.e. $|x(t)| \leq M, M < \infty$, then output is bounded under certain conditions.

⚡ Condition for bounded output is

$$|y(t)| = \left| \int_{-\infty}^{\infty} x(t-\tau)h(\tau)d\tau \right| \leq \int_{-\infty}^{\infty} |x(t-\tau)||h(\tau)|d\tau \leq M \int_{-\infty}^{\infty} |h(\tau)|d\tau \leq MK < \infty$$

where the integral $\int_{-\infty}^{\infty} |h(\tau)|d\tau \leq K < \infty$
i.e., the impulse is absolutely integrable.

To Calculate if an LTI system is BIBO Stable

- ⚡ Compute its impulse response $h(t)$.
- ⚡ Integrate it over $-\infty$ to ∞ , effectively over the given support of the signal.
- ⚡ If the integration is finite, the system is BIBO Stable.

Classwork

Classwork

Classwork

Classwork

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Up Next

⚡ Fourier Series